
What a Trained Recurrent Memory Obeys: a Mode-Mass Budget for Retention, Latching, and Forgetting

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Abstract

1 A physics-structured recurrent unit advances its latent state (q, p) by a damped
2 symplectic velocity-Verlet step of a learned Hamiltonian, and is built to remember.
3 What a *trained* one remembers, of what, and for how long has had no closed-
4 form account. We supply one: a mode-mass budget, verified on trained models
5 at latent dimension 4 on an S^1 testbed, ≤ 5 seeds, laptop CPU. Retention half-
6 life scales as $n_{1/2} \propto \mu^{-2}$ in the spectral mass μ — a Gell-Mann–Oakes–Renner
7 law, exact to $1.000000 \pm 5 \times 10^{-12}$ over 4.5 decades — and saturates at a mass-
8 independent floor past an exceptional-point crossover at $\varepsilon\mu \approx \gamma/2$. The trained
9 potential’s symmetry realization sets μ^2 ; the map’s (ε, γ, T) sets what μ^2 buys.
10 Three consequences carry the paper. (i) A recently posted machine-learning lifetime
11 law is the budget’s *overdamped face*: run unchanged on our trained units it matches
12 to $\leq 3\%$, median-consistent with its reported 1.013, and mispredicts past the
13 crossover in a calculable direction. (ii) A condensate-resolving linear spurion on
14 8 trained $SO(2)$ checkpoints makes the three chiral-perturbation-theory objects
15 separately measurable and satisfies GMOR proper, $\mu^2 F^2 = \delta\Sigma$. (iii) Designed
16 protection beats emergent protection by 13–14 orders of magnitude in μ^2 , the
17 retention triad is qualitatively absent in well-trained LSTM/LEM baselines, and
18 the prior is priced: a param-matched free twin fits $\approx 15\times$ better at short horizons,
19 then diverges past a measured ≈ 700 -step crossing while the unit stays bounded.
20 Designed-testbed results are labeled verification; learned-system results, evidence.

21 1 Introduction

22 A structure-preserving alternative to gated recurrence replaces the black-box update with a physical
23 integrator: a latent state (q, p) advanced by a symplectic velocity-Verlet step of a learnable separable
24 Hamiltonian $H(q, p) = T(p) + V_\theta(q)$, with an optional per-step damping $p \mapsto (1 - \gamma)p$ supplying
25 controllable forgetting. Such a unit is built to remember; how much, of what, and for how long has
26 had no closed-form account on a trained model. This paper supplies one — not a new architecture,
27 but a closed-form account of retention, verified on trained models and compared head to head with a
28 recently posted machine-learning lifetime law and with well-trained baselines. The reference unit is
29 the CLU (Causal Learning Unit), introduced as CHLU in Jawahar & Pierini (2026); results hold for
30 the class of damped symplectic recurrences, and the exactly-solvable theory verified here is developed
31 in a separate note, cited as *the theory note* (Anonymous, 2026, in the supplementary material).

32 **Nomenclature, not to be conflated.** Two quantities are called “mass” and run in opposite directions.
33 The inertial mass M_i , the learned diagonal of T : larger M , slower. The spectral mass μ_k , with
34 $\mu_k^2 = \lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$ at a critical point: larger μ , shorter memory. All retention statements
35 use μ .

Contributions. (1) The mode-mass budget, verified on trained models (Appendix C): the μ^{-2} law at overdamped slope -0.985 against a predicted -1 , the spectral-mass identity to 5×10^{-12} over 4.5 decades, the mass-independent floor and the exceptional point's $\sqrt{h} - h^*$ onset on 5 designed seeds at dim 4, and GMOR proper on 8 trained $SO(2)$ checkpoints. (2) The head-to-head against Mo's lifetime law on trained checkpoints (§4.1), reproduced on Mo's own estimator. (3) The symmetry-realization taxonomy exhibited by trained models (§3), with the designed-versus-emergent gap and the price of the physics (§4.2). (4) The absence of the retention triad in learned baselines, and the recipe that keeps a designed vacuum broken (§4.3). Designed-testbed results — invariant potentials, analytic tilts and spurions — are verification of the theory's exactness; learned-system, training-dynamics and head-to-head results are evidence.

2 Related work

Neutrality of group-orbit directions is classical (Golubitsky, Stewart & Schaeffer 1988; Krupa 1990; Rumberger 2001). Closest to us, Mo (2026) proves that a C^1 field exactly equivariant under a Lie group has at least $\dim(G/\mathcal{H})$ zero Lyapunov exponents tangent to the orbit; we use his diagnostics and his breaking-and-censoring protocol. His result is the *kinematics* of protection, silent on what sets the gap; here the gap is *constitutive*, and Lyapunov spectra are blind to the write semantics — transport, latch, noise accumulation — that make a neutral direction a *memory*. His theorem states sufficiency; equivariance is not necessary, the latch being a modulus of V_θ . Constructive equivariant memory (Di Bernardo et al. 2025; Keller 2025; Iqbal, Keller, Song, Miyato & Welling 2026) engineers *which* symmetry a network expresses but does not price *time*: we claim no novelty for choosing a symmetry group or coset, only the price list attached to a chosen structure. Our comparators are LSTM (Hochreiter & Schmidhuber 1997), LEM (Rusch et al. 2022) and coRNN (Rusch & Mishra 2021a); conformal symplecticity is McLachlan & Perlmutter (2001), the leapfrog $h < 2$ limit standard (Hairer, Lubich & Wanner 2006), and wake-sleep (Hinton et al. 1995) with PCD (Tieleman 2008) the contrastive objective, whose landscape distortion is classical (Fischer & Igel 2010; Nijkamp et al. 2020).

Retention guarantees elsewhere, and what we therefore do not claim. They are not this family's exclusive property, and the neighbours are occupied ground we cite rather than claim. HiPPO-LegS (Gu et al. 2020) has an $O(tL/\sqrt{N})$ approximation bound, a polynomial $\Theta(1/t)$ rather than exponential gradient decay and *exact* timescale equivariance with no discretization step to tune, and we have no measurement saying a learned mass spectrum is preferable to it; location-addressable retrieval of dynamical attractors is published (Kong, Brewer & Lai 2024); the NTM/DNC lineage was already fully soft, with failure modes that transfer (Graves et al. 2014, 2016; Csordás & Schmidhuber 2019); tunable permanence is occupied (EDEN, Karuvally et al. 2025); UnICORNN (Rusch & Mishra 2021b) is Hamiltonian, structure-preserving and *has* the gradient-bound theorem we lack; and the modern Hopfield update *coincides with* attention (Ramsauer et al. 2021), for a single head with identity projections and $\beta = 1/\sqrt{D}$. Our claim is correspondingly narrow: a physics-structured associative memory in which retention is a designed, per-item property — permanent (symmetry-protected, $\mu^2 \equiv 0$ exactly) or of a finite half-life set by μ^2 — rather than an emergent consequence of a learned recurrence — *controllability*, which we evidence, not *capability*, which we do not claim against transformers or state-space models. The nearest published neighbour to our write is Titans (Behrouz et al. 2025), test-time gradient descent with momentum on an associative-memory loss: the same family of discretized damped second-order dynamics, differing in that our damping and inertia are *read out and priced*, not optimizer hyperparameters.

Four claims we retire, so that a reviewer's first objections are answered in writing. (1) “*Attention and state-space models have no retention-guarantee analogue.*” They do; HiPPO-LegS above is the counterexample, exact where ours is a discrete map at fixed ε with $h = \varepsilon\mu < 2$ a hard stability limit. (2) “*Retrieval is a rollout, not a weighted sum*” as our novelty. The move is Kong, Brewer & Lai's, at capacity $N_c \propto K^{1.08 \pm 0.01}$ for their one-hot and binary index codings and 1.17 ± 0.02 for their 2D coding; our differences are structural — neither symplectic, Hamiltonian nor mass-structured there, and the address enters through an index channel rather than as the initial condition of the state. (3) “*Continuity is our escape hatch from the NTM/DNC lineage.*” False — both were already fully soft, so the precedent transfers as a warning: of the three DNC failure modes Csordás & Schmidhuber diagnose, key/value entanglement, which they rank first, is architectural in a memory whose address q_0 is both the key and the initial condition of the value; de-allocation aliasing maps onto any position-gated forgetting; chained-read degradation is the third; none are measured here. (4) “*A competitor to*

91 *the transformer.*” Never claimed, and indefensible at this scale: ≈ 1 –1.6 bits per emergent register, no win over
 92 a trivial baseline on any external benchmark, and formal limits on fixed-size latent state for copying and retrieval
 93 (Jelassi et al. 2024) that conservation does not evade — it prevents *decay*, not *capacity*.

94 3 Setup, and the two axes

95 **The map and the budget.** One dissipative velocity-Verlet step with step ε and damping γ advances
 96 (q, p) by a kick–drift–kick of $H = T(p) + V_\theta(q)$ followed by $p \mapsto (1 - \gamma)p$. It is conformally
 97 symplectic, $J^\top \Omega J = (1 - \gamma)\Omega$ with $\det J = (1 - \gamma)^d$, so memory at each normal mode reduces to one
 98 2×2 block ($\mu^2 = k/m$, $h = \varepsilon\mu$) with three bands: a latch ($\mu = 0$, symmetry-protected) with frozen
 99 displacement $q_\infty = q_0 + \varepsilon p_0/(M\gamma)$ and infinite half-life; an overdamped register ($0 < \varepsilon\mu \lesssim \gamma/2$)
 100 at $n_{1/2} \approx 2\gamma \ln 2/[(2 - \gamma)(\varepsilon\mu)^2]$; and an underdamped working memory ($\gamma/2 \lesssim \varepsilon\mu < 2$) at the
 101 mass-independent envelope half-life $2 \ln 2/(-\ln(1 - \gamma))$. The boundary is the exceptional point
 102 $h^* \approx \gamma/2$. Trained models are the reference unit’s Experiment-D configuration — dim 4, hidden 64,
 103 learned-Newtonian kinetic term, $dt = 0.05$, unit circle at 256 points, 150 epochs (§4.3 for why not
 104 the default 1000) — in two families: designed, an architecturally- $SO(2)$ -invariant V_θ on 5 seeds, and
 105 emergent, a generic MLP V_θ on the same data on 3 seeds.

106 **Axis 1: the symmetry realization of the trained potential.** It fixes each mode’s μ^2 , a statement
 107 about V_θ alone, in three cells, each realized by a trained model here. *Wigner–Weyl*: unbroken,
 108 invariant vacuum, $\mu^2 > 0$ degenerate within each irrep by Schur — the contrastive-divergence-
 109 eroded vacuum of §4.3. *Nambu–Goldstone*: spontaneously broken, vacuum on a G -orbit, $\mu^2 = 0$
 110 exactly along $\dim(G/\mathcal{H})$ directions — the designed unit. *Pseudo-Goldstone*: explicitly broken, tilted
 111 orbit, $\mu^2 = \delta\Sigma/F^2$ small — analytic tilts and the self-broken MLP. Only the *classical, tree-level*
 112 Goldstone theorem is used, a statement about a Hessian: if V_θ is G -invariant and its minimiser is not,
 113 the orbit consists of minimisers and $\nabla^2 V_\theta(q^*)$ annihilates every orbit tangent. No thermodynamic
 114 limit, no phase transition, no $\hbar$, and no loops, chiral logarithms, running low-energy constants or
 115 anomalies anywhere in this paper. *Axis 2* is the map: (ε, γ, T) fixes what a given μ^2 buys, through
 116 the three bands above.

117 *Fine print, next to the claims it qualifies.* (a) The latch’s infinite half-life is a $T = 0$ statement about the
 118 deterministic damped map; at $T > 0$ the coset coordinate diffuses with $D_\theta = \varepsilon T(2 - \gamma)/(2F^2\gamma)$, $F^2 =$
 119 $M_{\text{ch}} r^{*2}$ — a *diffusive register* of finite computable lifetime, verified on the trained designed checkpoints, where
 120 raising friction *lengthens* coset memory by $3.77 \pm 0.23\times$ for $\gamma : 0.05 \rightarrow 0.2$ on 5/5 seeds, and holding
 121 only under fluctuation–dissipation-consistent noise and a Newtonian kinetic mode, never under the reference
 122 implementation’s legacy noise default. All §4 results are $T = 0$. (b) By Sylvester’s law of inertia a flat direction
 123 keeps $\mu^2 = 0$ under any inertial anisotropy: symmetry protects the write *current*, flatness protects the *register*,
 124 and map equivariance is sufficient but not necessary for a neutral memory direction — our kinetically-broken
 125 battery is a *measured* non-equivariant map that latches exactly (Appendix B). (c) The *tilt-as-a-designed-lifetime-*
 126 *dial* construction is claimed for designed geometries only, the self-broken MLP being an *observed* instance of the
 127 cell and not a dial: in a separate battery of ours on a *learned, multi-atom* store, a breaking coefficient intended as
 128 the lifetime dial moved the softest eigenvalue *the wrong way*, and a coercivity term — not the breaking — set
 129 the ceiling (Appendix B).

130 *Mandatory flag, travels with every finite-temperature number in this paper.* All such results require
 131 `langevin_noise="fdt"` — the fluctuation–dissipation-consistent noise scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$ — and
 132 a Newtonian kinetic mode (`newtonian_identity/newtonian_learned`). The reference implementation’s
 133 default is `legacy`, under which T is not in energy units and none of these laws hold; and in the relativistic
 134 kinetic mode no noise scale targets Gibbs at all (Appendix B; *theory note*, Prop-9’), so the laws are undefined
 135 there. The checkpoints here are `newtonian_learned`, so every finite-temperature number here is in scope.
 136 Nothing in §4 depends on it: all main-text retention results are $T = 0$ deterministic rollouts.

137 4 Results

138 Every result below lives on a trained or learned system. The budget’s exactness on the designed
 139 testbed is verification, reported in Appendix C: constitutive identity to 3.2×10^{-10} over all 70 tilt
 140 rows, GMOR ratio $1.000000 \pm 5 \times 10^{-12}$ across 4.5 decades, overdamped slope -0.985 against a
 141 predicted -1 , floor 27.03 steps at $\gamma = 0.05$, onset exponent 0.5165 against a predicted $1/2$, and
 142 GMOR proper to 1.33×10^{-15} absolute on 8 trained checkpoints. With that 2×2 block established,
 143 to machine precision, as the trained map’s own angular block, the sections below ask what it buys.

Mo's exact lifetime protocol run on trained CLUs
overdamped: 1.012–1.03 (his median 1.013) · EP: 2.20 ± 0.16 · deep underdamped: 0.31

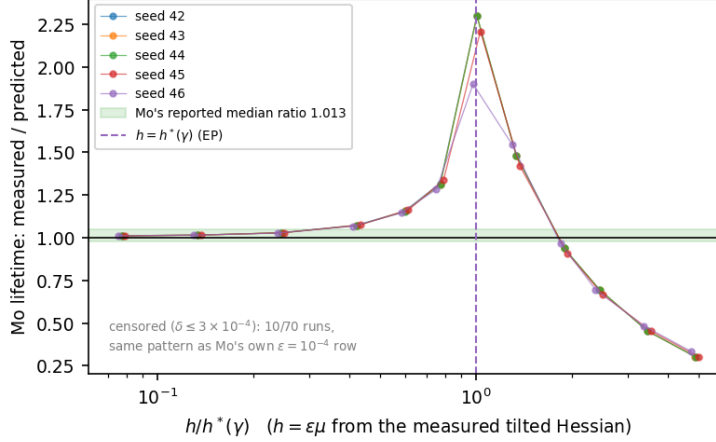


Figure 1: Mo’s single-exponential lifetime law, run *unchanged* on the trained models, is the overdamped face of the mode-mass budget. Measured/predicted lifetime ratio versus breaking magnitude, 14 regimes, 5 seeds: ≈ 1 overdamped, reproducing his reported median 1.013; a $2.2\times$ delay spike at the exceptional point; declining to $0.31\times$ deep underdamped — containment exact below the crossover, failing above it in the calculable ballistic direction. *Evidence*.

144 4.1 Head-to-head against a recently posted lifetime law, on trained models

145 *Evidence: a recently posted ML lifetime law, run on trained checkpoints.* Mo (2026) proves that
146 an exactly equivariant recurrent flow has at least $\dim(G/\mathcal{H})$ zero Lyapunov exponents along the
147 group orbit — a lower bound — and shows that a “pseudo-gap” predicts a finite memory lifetime
148 through a single-exponential estimator: the kinematics of protection, silent on what *sets* the gap.
149 Running his exact code-level protocol — phase 0.35 rad, threshold 0.2 rad, his censoring, cap 15000
150 steps — unchanged on the 5 trained models across 14 breaking magnitudes gives measured/predicted
151 $1.012 \pm 0.000 \rightarrow 1.029 \pm 0.001$ overdamped against his reported median 1.013, 2.202 ± 0.155 at
152 the exceptional point, and 0.309 ± 0.012 deep underdamped. Mo’s law is the overdamped face of
153 our budget: in his regime the two coincide, $\text{corr}(\log \text{pred}, \log \text{meas}) = 0.9987$ overdamped-only,
154 and his predictor fails only past the crossover — by $\approx 3.2\times$ at the deepest trained-model breaking
155 we test, $\delta = 4$, our sweep going no deeper underdamped — in the calculable ballistic direction,
156 with the $2.2\times$ exceptional-point delay spike as the third signature. On the *exact analytic map* the
157 same mechanism continues to $\approx 5\times$ further underdamped (theory note, check (k), $\gamma = 0.1$); the
158 $\approx 3.2\times$ is the trained-model number, the $\approx 5\times$ is exact-map only, and we never quote it as evidence
159 about a trained network. This is containment, not conflict: below h^* we recover his reported number;
160 above it the regime structure — floor, ringing, exceptional point — is what a first-order flow cannot
161 exhibit and what a *constitutive* gap adds. Nor does it rest on substituting the exact Jacobian gap for
162 his predictor: with Mo’s own finite-horizon estimator $\hat{\lambda}(T=128)$ the result holds on his instrument,
163 $\text{corr} = 0.9995$ overdamped, meas/pred 0.86–1.03, and 0.30 at $\delta = 4$ against the exact-gap 0.31.

164 4.2 Designed versus emergent protection, and the price of the physics

165 A generic MLP potential trained on perfectly symmetric data develops *no* near-flat direction: softest
166 emergent $\mu^2 = 5.1/5.9/5.4 \times 10^{-2}$ on 3 seeds against the designed flat mode $\leq 2.4 \times 10^{-15}$, a gap of
167 13–14 orders; Mo’s own attribution instrument separates the architectures by 15 orders. Operationally
168 it stores nothing on a *continuum*: ≈ 1 –1.6 bits on 2–3 washboard minima, a written δ relaxing
169 completely — though the damping-optimum retention law itself still generalizes to that arm.

170 **The price of the prior (evidence).** Against param-matched controls — dim 4, 150 epochs, $\pm 0.05\%$
171 params, 3 seeds, same MLP potential — the picture is Pareto, not podium. Volume conservation buys
172 bounded (BIBO) dynamics within the coercive-potential / compact-sublevel-set scope (1.00 versus

0.33 for a broken-volume arm), a protected near-flat direction (μ^2 0.008 versus 0.122) and vacuum survival under contrastive-divergence training ($r^* = 0.72$ versus 0); leapfrog buys the latch; and the prior costs raw fit, an unconstrained twin fitting $\approx 15\times$ better, 0.0128 against 0.190 eval MSE. That cost is *contraction-forbidden* by volume conservation — the broken-volume arm recovers $\approx 2.4\times$ of it precisely by leaking volume — and is not the causal velocity cap, inactive here. The loan is called at ≈ 700 steps (measured): the twin leads by 1–2 orders to ≈ 500 steps, crosses at ≈ 700 and diverges to 196 at 5000, while the unit holds a bounded plateau ≈ 0.20 – 0.23 . Its asset is boundedness by construction, not the lowest steady MSE — broken-volume (0.14) and LSTM (0.13) plateau below it — and of the licensed repairs only global damping buys the fit back, a learned friction *field* does not (Appendix A.1, with the cross-experiment non-comparability caveat its numbers require): paid mechanisms are not interchangeable.

4.3 Learned baselines, and the recipe that keeps the vacuum broken

Evidence. On Mo’s S^1 path-integration family the baselines are trained with a learning-rate sweep, best RMSE kept, and are not strawmen: LSTM and LEM reach train-horizon RMSE 0.18/0.23 rad. Yet under Mo’s *autonomous* protocol the structural triad of latch, μ^{-2} law and bounded drift is qualitatively absent in them: coRNN/LEM/LSTM forget the stored analog phase in $\approx 5.6/56/69$ map-steps, drift randomizes past 1.2 rad, and they are *perturbation-fragile*, a 0.1 hidden kick collapsing LSTM $69 \rightarrow 2$ and LEM $56 \rightarrow 5$, while the generically-trained unit holds ≈ 263 map-steps with bounded drift (≈ 0.35 rad, budget law to $+12$ – 15%) and the designed unit never moves (5/5 seeds).

Honest gap. The load-bearing claim is the qualitative triad, architecture- and compute-independent. The raw 263-versus-69 map-step ratio does not survive compute normalization, and we retire it as a compute claim: per step the Verlet update costs $\approx 6.2\times$ an LSTM cell and $\approx 3.1\times$ a LEM cell in wall time, ≈ 14 – $15\times$ FLOPs, not width-matched, so per unit of retention the unit spends $\approx 23.5\times/\approx 14.6\times$ more wall time than the baselines it out-holds (Appendix A.2). Lacking native velocity ingestion, it cannot enter Mo’s input-driven task-RMSE axis at all, so no task-RMSE was fabricated.

The designed vacuum also needs a training recipe, because the default one destroys it. Wake-sleep contrastive divergence drives the order parameter r^* to zero, inverting the designed vacuum in 8/8 runs at the default 1000 epochs: a symmetry-restoration (condensate-melting) transition in the sense of §3, the potential landing in the Wigner–Weyl cell with the predicted degenerate μ^2 pair, on a horizon set by sleep-update *frequency* racing the wake clamp, and specific to flat/degenerate vacua. *Scope clause, travelling with that finding:* our sweep varies chain length only over `sleep_steps` $\in \{50, 500\}$ at otherwise fixed sampler settings and does not resolve where either sits relative to this model’s mixing time, so the finding is *frequency-decisive across the two chain lengths we run*, and is not a claim that chain length is irrelevant to contrastive-divergence learning in general.

A $V(\text{data})$ energy anchor holds the condensate up — $\lambda = 100$: 5/5 seeds, $r^* = 0.911 \pm 0.016$, at the cost of weaker noise rejection and $\approx 35\times$ higher wake MSE — and every headline law of the budget still holds under it at $\approx 20\times$ the erosion horizon (Figure 2): 3000 anchored epochs, 3 seeds, GMOR exact to 1.5×10^{-12} over 4.6 decades, slope -0.956 , same floor, exceptional-point onset bit-identical at 0.5165. The architecture *breaks* the symmetry, the objective *restores* it, the anchor *keeps it broken*.

5 Discussion

Scope box — limitations and scope choices, stated so no reader has to infer them. All results are laptop-CPU, dim 4, the S^1 testbed, ≤ 5 seeds; the solvable core is quadratic and trained statements are local at a critical point of V_θ , with anharmonic deviations measured as a zero-kick retention bias of $+5$ to $+29\%$, kick-independent for 2/3 emergent seeds and amplitude-dependent for the softest- μ^2 seed. The head-to-head is on *autonomous retention* in map-application units, and the unit does not enter the input-driven task-RMSE axis. BIBO boundedness is evidenced within the coercive-potential / compact-sublevel-set scope and concerns the autonomous, designed setting measured here — nothing here claims that symplectic structure buys off-distribution stability in a *trained, input-driven* setting, which we do not test and where a published symplectic architecture already holds the ground with a gradient bound — and friction never stabilizes a saddle (Appendix B). The symmetry framing is tree-level only. The finite- T diffusion law holds on both the designed channel (verification) and the emergent arm (evidence, 3 seeds, above $T^* \approx 3 \times 10^{-3}$) under FDT-consistent noise in a Newtonian kinetic mode; the one face that does not generalize is the continuous latch register, which is designed-only. (i) *Scale is out of scope*

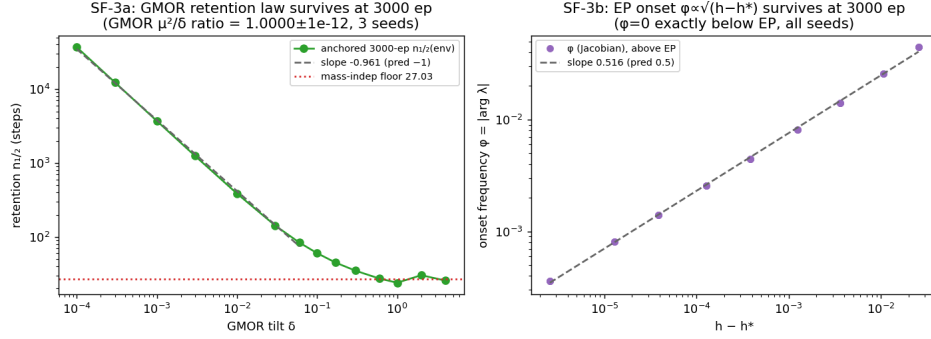


Figure 2: The headline laws of the budget survive the shipped anchor cure at 3000 epochs ($\approx 20 \times$ the erosion horizon; anchored $\lambda=100$, 3 seeds). Left: the GMOR retention law $n_{1/2}$ versus tilt δ — anchored-3000-epoch points on the predicted slope, fitted -0.961 against predicted -1 , saturating at the mass-independent floor 27.03; the μ^2/δ ratio is 1.0000 ± 10^{-12} over 4.6 decades. Right: the exceptional-point onset $\varphi \propto \sqrt{h - h^*}$ above the exceptional point, fitted slope 0.516 against predicted 0.5, with $\varphi = 0$ exactly below — bit-identical to the 150-epoch value. *Designed tested under the shipped cure; the survival of the laws through 3000 erosive epochs is evidence for the recipe.*

226 *by choice*: every number here is dim 4, hidden 64, ≤ 5 seeds, single-unit, laptop-CPU, and we report no result
 227 at larger dimension, wider layers, more units or accelerator scale. (ii) *No external benchmark is won on its own*
 228 *headline metric anywhere in this paper*; the comparisons are diagnostic retention protocols on a synthetic S^1
 229 family and matched-parameter ablations, not leaderboard results — and §4.3’s honest gap is part of the claim,
 230 not a caveat to it. (iii) *The substrate scope, once, in our own voice: these laws govern the store; end-to-end*
 231 *performance additionally depends on the encoder; measured separately, with its parameter and state-byte budget*
 232 *ledgered.*

233 The budget suggests reading a trained memory as an *effective field theory of retention*: latches as exact
 234 flat directions, the moduli of V_θ ; registers as pseudo-Goldstone modes of a chosen half-life, fenced
 235 as designed-geometry in Appendix B; kinetic isotropy as the price of an equivariant write *current*,
 236 not of the register. Named as directions, not results: the μ^{-2} law, crossover and floor re-measured at
 237 latent dimension $\gtrsim 64$ and $N \geq 8$ interacting units, where mode mixing can break the single-block
 238 reduction that makes the budget exact here; the head-to-head on a wider trained-model family; the
 239 anchored recipe at longer training and larger width; the equivariant-control wrapper that would put
 240 the unit on the supervised task axis; a learned-store re-test of the explicitly-broken cell respecting
 241 that fence; the (μ, γ, T) budget cube whose $T = 0$ face is §4; and the price list re-derived beyond
 242 abelian $SO(2)$, for a torus T^2 or the non-abelian $SO(3) \rightarrow SO(2)$ coset — a direction, on which
 243 this paper offers no evidence.

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A Supplementary results

A.1 The loan curve and the recovery ladder

Dim 4, hidden 64, learned-Newtonian kinetic term, dt = 0.05, circle R=1×256, 150 epochs, 3 seeds, wake-only objective except the γ_ϕ rung. Cumulative fit MSE against horizon on the stationary circle-vacuum orbit, param-matched to $\pm 0.2\%$ (CLU 4549 / broken-volume 4557 / twin 4551).

horizon (steps)	twin	CLU	broken-volume	LSTM
10	2.9×10^{-6}	2.2×10^{-5}	1.1×10^{-4}	7.5×10^{-2}
50	6.2×10^{-5}	1.2×10^{-3}	1.0×10^{-3}	4.9×10^{-2}
100	2.4×10^{-4}	1.2×10^{-2}	4.1×10^{-3}	4.9×10^{-2}
500	1.1×10^{-2}	1.8×10^{-1}	9.7×10^{-2}	7.7×10^{-2}
1000	3.2×10^{-1}	2.0×10^{-1}	1.2×10^{-1}	9.9×10^{-2}
5000	1.96×10^2	2.2×10^{-1}	1.4×10^{-1}	1.3×10^{-1}

The twin is best by 1–2 orders to ≈ 500 steps, crosses above CLU at ≈ 700 and diverges to 196 at 5000; CLU is the only arm flat *by construction*, on a bounded plateau ≈ 0.20 – 0.23 . Broken-volume (0.14) and LSTM (0.13) plateau slightly *below* CLU (0.22): “physics costs raw fit” persists as a small steady-state penalty, and CLU’s asset is boundedness, not the lowest plateau.

Non-comparability caveat, to be read before using these numbers. This ladder is a **separate wake-only experiment**, and its **absolute MSE scale and its twin (0.0047) are not cross-comparable** with the param-matched table of §4.2 (twin 0.0128, different seeds, different training objective). **Only within-table gaps are meaningful.** In particular, do not divide across tables: the “ $\approx 15\times$ ” of §4.2 (0.190/0.0128) and this table’s implicit “ $\approx 44\times$ ” (0.2066/0.0047) are the *same qualitative statement* measured on two different scales, not two different findings, and the 92% below is a fraction of **this** table’s absolute gap (0.202) only. The $\gamma = 0$ conservative rung (0.2066) is this experiment’s counterpart of the $\gamma = 0$ CLU of §4.2 (0.190).

rung	eval MSE	% gap recovered	BIBO	latch drift	flat μ^2
twin (no physics)	0.0047	100% (def.)	1.00	—	—
CLU conservative ($\gamma = 0$)	0.2066	0% (def.)	1.00	0.778	5.2×10^{-3}
+γ global ($\gamma = 0.05$)	0.0216	92%	1.00	0.778	5.2×10^{-3}
+γ_ϕ learned field ($K = 2$)	0.2548	−24%	1.00	0.706	2.1×10^{-2}

Each licensed mechanism is added one at a time; the gap recovered is $(\text{CLU}_{\gamma=0} - \text{rung}) / (\text{CLU}_{\gamma=0} - \text{twin})$, twin gap 0.202. Global γ recovers 92% of the **absolute** gap at zero structural cost, leaving BIBO, latch and μ^2 unchanged, while still trailing this table’s twin by $\approx 4.6\times$ in *ratio* (0.0216 against 0.0047); both statements must travel with the claim. The learned γ_ϕ field is a targeted-forgetting tool and does not recover fit, at -24% ; that rung retrained with the sleep phase on, so the “wrong tool” verdict is robust but the -24% is not a clean single-knob delta.

A.2 Per-step compute: CLU-Verlet against LSTM and LEM

Single-core CPU, f32, XLA; this backs the retirement of the “ $\approx 4\times$ longer” as a compute claim in §4.3. One CLU dissipative-Verlet step (kick–drift–kick, two $\nabla_q V$ backprops through a dim-4 $\rightarrow$ 64 $\rightarrow$ 64 $\rightarrow$ 1 tanh MLP, learned-Newtonian kinetic term, $\gamma=0.05$, $dt=0.05$) is measured against one LSTM and one LEM cell, both input 1, hidden 16, output 2, the exact retention baselines of §4.3. Wall time is the median of 7 scan-amortized repetitions over 2×10^5 steps; the naïve single-call timing is dispatch-bound at a $\approx 5 \mu\text{s}$ floor and is *not* reported.

unit	params	FLOPs/step	wall ns/step (scan-amortized)
CLU Verlet (h64, dim 4, KDK)	4549	36148	1554
LSTM cell (h16)	1186	2512	252
LEM cell (h16)	1186	2400	500

normalization	CLU vs. LSTM	CLU vs. LEM
raw map-steps (263 vs. 69/56)	$3.8\times$ longer	$4.7\times$ longer
FLOP-normalized	54.8$\times$ more	70.7$\times$ more
wall-normalized	23.5$\times$ more	14.6$\times$ more

Per-step ratios: CLU/LSTM = $14.4\times$ FLOPs and $6.2\times$ wall; CLU/LEM = $15.1\times$ FLOPs and $3.1\times$ wall. The $4\times$ **inverts** under compute normalization: per unit of retention the unit spends ≈ 15 – $24\times$ the wall time and ≈ 55 – $71\times$ the FLOPs of the RNN it out-holds. **Confound, flagged:** the comparison is not width-matched, CLU at hidden 64 with two gradient evaluations against baselines at hidden 16; a width-matched CLU would narrow the gap, but the sign is robust because the Verlet step requires two potential backprops, and the 263/69/56 retention numbers come from these specific configurations. The measurement is single-core CPU at batch 1; GPU or batched throughput could shift the wall ratios, though not the FLOPs. §4.3 accordingly leads with the qualitative triad, states the per-step $\approx 6\times$ (LSTM) and $\approx 3\times$ (LEM) wall factor, and does not carry the map-step ratio as a compute claim.

B Prominent negatives

Negatives are recorded, never dropped. Each row states the claim tested, the result, and the number that settles it.

claim tested	result	number
Does a learned inertial mass isotropize on symmetric data?	No; and the breaking is invisible to the Hessian and to the latch, showing only as a bounded Noether-charge oscillation.	Log-mass split ratios 0.98 / 1.18 / 5.4 from init to 150 ep; charge amplitudes 5.4/1.6/0.082 $\times 10^{-2}$; tied controls exactly 1.0000.
Does wake-sleep contrastive divergence preserve a designed degenerate vacuum?	No — it inverts it. The cure is a $V(\text{data})$ energy anchor, and the headline laws survive it (§4.3).	$r^* \rightarrow 0$ in 8/8 runs at 1000 ep; inversion at epoch 116/442/959 by sleep frequency; anchor $\lambda=100$ holds 5/5 seeds at $r^*=0.911 \pm 0.016$.
Does an emergent (MLP) unit have a continuous coset register?	No — the continuous register is designed-only; an emergent unit stores a few bits on discrete washboard minima (evidence, 3 seeds; §4.2).	$1 - \lambda_{\text{coset}} \approx 10^{-3}$ against the designed $\leq 1.1 \times 10^{-15}$, ~ 12 orders; written δ retained $\leq 2.1 \times 10^{-3}$; capacity ≈ 1 –1.6 bits.
346 Do raw $n_{1/2}$ exponents discriminate designed from emergent?	No — instrument, not physics: a matched designed control on the same grid gives the same slopes, both carrying the ℓ_θ/Δ boundary layer. No $n_{1/2}$ may be quoted without its Δ and ℓ_θ/Δ .	Designed control $d \log n_{1/2}/d \log T = -0.53, -0.60, -1.04$ and $d \log n_{1/2}/d \log \gamma = +0.78, +0.63, +0.55$; ℓ_θ/Δ up to 2.0.
Is an explicit-breaking coefficient a lifetime dial on a learned store?	No, and the first half fails in sign: raising the breaking <i>lowers</i> the softest eigenvalue, and a coercivity term, not the breaking, sets the ceiling. A designed degeneracy does not survive superposition.	$\lambda_{\min}$ runs $+0.0994 \rightarrow -8.2846$ and $+0.3291 \rightarrow -1.1980$ over four decades of breaking; hard ceiling $\tau_{\max} = \Gamma/2\alpha = 4.0$; tilt-vacuum residual 0.140–0.343 against a random-direction baseline $1/\text{dim} = 0.167$; tangential curvature predicted 0.100, measured 0.0994.
Can this unit enter the input-driven path-integration task-RMSE axis?	No — it has no native velocity ingestion and the equivariant-control wrapper is unbuilt. No task-RMSE was fabricated.	—
claim tested	result	number
Do a learned friction field and a curvature governor compose usefully?	No — the pair is Pareto-dominated by either alone, and a learned compact-support gate under-covers without accurate placement.	Locus discovery 2/6 before adaptive- K , 2/3 after; best learned rejection 0.861; a λ re-sweep does not approach the oracle.
347 Can friction stabilize a saddle, or an energy gate catch isoenergetic escape?	No to both — a genuine scope limitation of the class. Only curvature control of V_θ and the mass-anisotropic cap $v_i^{\max} = c/\sqrt{M_i}$ bound expanding or isoenergetic modes.	$\lambda_+ > 1$ for all γ , with $\min_\gamma(\lambda_+ - 1) = 1.4 \times 10^{-5}$.
Is the coded momentum-coupled Langevin sampler FDT-exact in every kinetic mode?	No — exact only in the Newtonian modes; in the relativistic mode <i>no</i> noise scale gives the coded chain a Gibbs invariant at all.	Required per-mode scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$.
Is a mean-spectrum chaos regularizer informative?	No — it is degenerate; a chaos penalty must use a max or positive-part statistic.	—
Is <code>sleep_temperature</code> active at <code>sleep_friction=0</code> ?	No — a silent knob; cite it when reproducing temperature-swept sleep runs.	$\sigma \propto \sqrt{\gamma_{\text{sleep}}}$, so the swept runs are bit-identical.

348 **A reporting caution travels with the isotropization row.** The closed-form charge-oscillation amplitude is an
349 **on-vacuum-orbit** statement; **off the orbit only boundedness holds**. The amplitude-to-split ratio is moreover
350 **not** constant, as the exact envelope requires, since it is linear only to leading order — so no proportionality
351 constant should be quoted as a law, and we report amplitudes and boundedness, never a “drift rate”. The design
352 rule attached to mass-tying is correspondingly about the *current*, not the register: **an anisotropic channel still**
353 **latches with infinite half-life**; tie the masses for an angle-independent write current, and at $\dim(G/\mathcal{H}) \geq 2$ a
354 degenerate pseudo-Goldstone multiplet, **not** to save the register.

355 **Two reading rules travel with the learned-store row.** (a) The breaking coefficient of that battery is an **explicit-**
356 **symmetry-breaking magnitude**; it is **not** this paper’s integrator step ε , and the two must never be conflated. (b)
357 No sentence in this paper may be read as claiming that a tilt magnitude is a lifetime dial on a learned store, or
358 that a pseudo-Goldstone lifetime can be extended without bound by shrinking the breaking: on a store with a
359 coercive confinement $\alpha \|q\|^2$ the ceiling is $\Gamma/2\alpha$, and lowering α breaks the write. This paper’s own tilt results
360 (Appendix C) are analytic tilts on **designed** architecturally-invariant potentials, where the construction is exact;
361 the learned-store scope is exactly what this negative removes. That battery is a *different architecture* from the
362 trained units of this paper and shares none of their training configuration, which is why it is quoted as a scope
363 fence and never as a measurement of the units here. What survives it is the *geometry*: the identity $\lambda_{\min} = \epsilon$
364 holds in the single-atom geometry in which it was derived, unit-tested to 5%, and the registered damped-mode
365 floor $\tau = 2/\Gamma$ is confirmed, 5.0 predicted against 4.23–6.58 measured time units.

366 **The sampler row’s scope, stated precisely, because the qualifier is load-bearing.** The coded damping substep
367 is an additive Gaussian momentum kick of fixed covariance, so the chain’s invariant momentum marginal is
368 necessarily Gaussian-smoothed; for a *relativistic* kinetic term the Gibbs momentum marginal is Maxwell–Jüttner,
369 which is not of that form (*theory note*, Prop-9’ — a characteristic-function argument needing neither $V = 0$ nor
370 linear damping). **The failure is in the sampler, not in the thermodynamics**: the configurational equilibrium

371 $\pi_q \propto e^{-V_\theta/T}$ is relativity-insensitive, since the momentum integral factorizes out, so a relativistic unit's
372 equilibrium distribution is perfectly well-defined and the chain simply fails to sample it; a one-line thermostat
373 carrying a randomized, momentum-dependent inertia equal to the relativistic mass repairs it. We therefore never
374 assert that a relativistic unit “has no equilibrium”. **This touches no result in this paper:** the trained units here
375 are `newtonian_learned` throughout, so σ^* is exact for every configuration we run — the qualifier is a *scope*
376 *clause* on a class-level theorem, not a correction to a measurement.

377 C The mode-mass budget verified, and GMOR proper on trained checkpoints

378 *Designed testbed with analytic tilts and an analytic spurion, therefore verification of the theory's exactness;*
379 *5 seeds at 150 epochs plus 3 anchored at 3000 epochs, dim 4, laptop CPU, f64 probes, probe-only with no*
380 *retraining.*

381 **C.1 The budget, verified on trained models.** On the trained designed vacuum, whose flat mode has $\mu^2 \leq$
382 2.4×10^{-15} and is machine-flat, the constitutive identity is exact: the Hessian-derived spectral mass equals the
383 measured one-step-Jacobian gap to 3.2×10^{-10} over all 70 tilt rows. The GMOR law follows, $\mu_{\text{meas}}^2/\mu_{\text{pred}}^2 =$
384 $1.000000 \pm 5 \times 10^{-12}$ at every δ and seed across 4.5 decades, with fitted overdamped retention slope -0.985
385 ($n = 35$; predicted -1), saturating past the crossover at the mass-independent floor 27.03 steps at $\gamma = 0.05$.
386 Two second-order signatures confirm a genuine exceptional, defective point. *Retention-metric bifurcation:*
387 envelope half-life and first-crossing time agree to 0.01% overdamped and split by $3.2 \times$ at $\delta = 4$, so every
388 reported lifetime must name its metric. *Frequency onset:* $\varphi = 0$ exactly below h^* (15/15 rows, a real
389 eigenvalue pair) and $\varphi \propto \sqrt{h - h^*}$ above, holding down to $h - h^* = 2.6 \times 10^{-6}$ on a fine multiplicative
390 grid ($h - h^* \in [-4.2 \times 10^{-3}, 2.6 \times 10^{-2}]$), fitted slope 0.5165 against the predicted $1/2$ — the slight excess is
391 far-field bending — and prefactor 0.268 against the theory note's 2×2 -normalized 0.3247, a one-line units
392 reconciliation. Where the quality factor permits ($f = 2, 4$) the trajectory frequency matches the Jacobian
393 frequency to 0.06–0.3%; near onset the mode decays before one period completes, so the exceptional point is
394 spectroscopically real but dynamically silent there. The latch transport law $q_\infty = q_0 + \varepsilon p_0/(M\gamma)$ holds to
395 $\leq 1\%$ over two decades of γ , the written angle then frozen to $\leq 1.2 \times 10^{-15}$ rad.

396 **C.2 The damping corollary** — a supporting result, deliberately not a headline. The same block fixes the
397 half-life's dependence on the damping, and that dependence is **not** monotone: sweeping γ at fixed spectral mass,
398 $n_{1/2}(\gamma)$ traces a V-shape of depth $\approx 8 \times$ whose argmin sits at, or just below, the exact critical-damping value
399 $\gamma^* \approx 2\varepsilon\mu$, on all 5 designed seeds. Below γ^* the mode is underdamped at the floor $2 \ln 2/(-\ln(1 - \gamma))$, which
400 *shortens* as γ grows; above it the mode is overdamped at $n_{1/2} \approx 2\gamma \ln 2/[(2 - \gamma)(\varepsilon\mu)^2]$, which *lengthens*; the
401 exceptional point is where the branches meet. Two consequences: (i) a retention measurement quoted at one γ is
402 uninformative about the sign of $\partial n_{1/2}/\partial \gamma$ unless the band is named, the same architecture giving both signs;
403 (ii) the $\mu \rightarrow 0$ latch corner is the limit $\gamma^* \rightarrow 0$, so a Goldstone register is permanently overdamped, which is the
404 structural reason friction cannot erase it.

405 **C.3 GMOR proper: why a second spurion.** The angular tilt $V \rightarrow V + \delta(1 - \cos n\theta)$ of C.1 is *radius-*
406 *independent* — measured on these checkpoints, $\max |r_{\text{tilt}}^* - r^*| = 2.22 \times 10^{-15}$ over all (checkpoint, δ) pairs
407 — so it measures only the **product** $\mu^2 F^2 = \delta n^2$, in which the condensate degenerates to the pure number n^2
408 (measured: $\max |\mu^2 F^2/(\delta n^2) - 1| = 1.03 \times 10^{-11}$): the correct instrument for verifying a power law, and
409 provably incapable of resolving the condensate. The **linear ambient spurion** $V \rightarrow V - \delta(u \cdot q)$ — the exact
410 analogue of the ChPT quark-mass term, with u in the channel plane — lets the vacuum radius run with δ , so the
411 three GMOR objects become separately measurable on a trained checkpoint: the pseudo-Goldstone spectral mass
412 μ^2 from the autodiff Hessian at the polished vacuum; the decay constant $F^2 = M_{\text{ch}} r^{*2}$, the learned channel
413 inertia times the vacuum radius squared and **not** the orbit radius; and the condensate $\Sigma = r^*(\delta) = -\partial E_{\text{vac}}/\partial \delta$.
414 Over 8 trained checkpoints and $\delta \in [10^{-8}, 0.3]$ (10 values, 8 decades), $\max |\mu^2 F^2 - \delta \Sigma| = 1.33 \times 10^{-15}$ over
415 all 80 pairs.

δ	$\Sigma = r^*$ (mean)	$\mu^2 F^2$ (mean)	max rel. dev.	max abs. dev.
10^{-8}	0.95186937111604	$9.5186937286 \times 10^{-9}$	1.10×10^{-8}	1.01×10^{-16}
10^{-7}	0.95186948174437	$9.5186948223 \times 10^{-8}$	1.87×10^{-9}	1.85×10^{-16}
10^{-6}	0.95187058802597	$9.5187058802 \times 10^{-7}$	1.87×10^{-10}	1.86×10^{-16}
10^{-5}	0.95188165067351	$9.5188165068 \times 10^{-6}$	2.18×10^{-11}	2.16×10^{-16}
10^{-4}	0.95199226031843	$9.5199226032 \times 10^{-5}$	1.34×10^{-12}	1.33×10^{-16}
10^{-3}	0.95309668422658	$9.5309668423 \times 10^{-4}$	2.27×10^{-13}	2.09×10^{-16}
10^{-2}	0.96398369127653	$9.6398369128 \times 10^{-3}$	2.69×10^{-14}	2.69×10^{-16}
3×10^{-2}	0.98732513581811	$2.9619754075 \times 10^{-2}$	9.66×10^{-15}	2.71×10^{-16}
10^{-1}	1.06435252346294	$1.0643525235 \times 10^{-1}$	1.55×10^{-15}	1.53×10^{-16}
3×10^{-1}	1.60757655208590	$4.8227296563 \times 10^{-1}$	1.55×10^{-15}	1.33×10^{-15}

F-1: condensate-resolving GMOR on trained $SO(2)$ CLU checkpoints (probe-only, no retraining)

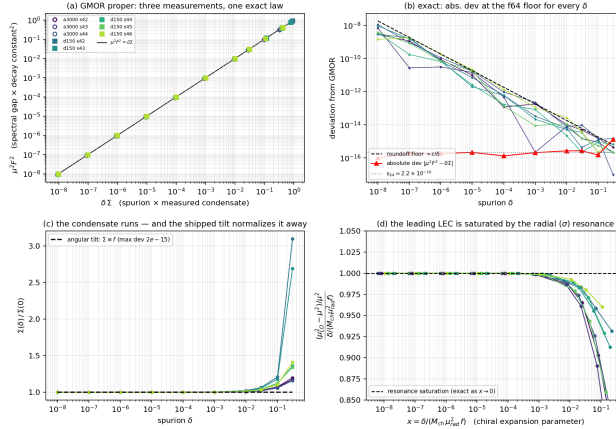


Figure 3: GMOR proper on trained checkpoints, with a linear ambient spurion. (a) The collapse $\mu^2 F^2 = \delta \Sigma$ over 8 decades of δ , 8 trained $SO(2)$ checkpoints. (b) Absolute deviation pinned at the f_{64} floor ($\leq 1.33 \times 10^{-15}$) against the ϵ/δ relative-roundoff line of the autodiff Hessian — the small- δ relative residual is a probe artifact, not a law violation. (c) The running condensate $\Sigma(\delta)/\Sigma(0)$ under the linear spurion against the angular tilt’s exactly flat $\Sigma \equiv r^*$ — the shipped tilt cannot see the condensate. (d) The low-energy-constant ratio tending to 1 as the chiral expansion parameter $x \rightarrow 0$, with slope -1.05 . *Designed testbed, analytic spurion, trained checkpoints, therefore verification.*

417 **Precision fine print, next to the claim.** GMOR holds to machine precision **in absolute terms at every**
418 δ ; relative exactness ($\leq 2.7 \times 10^{-14}$) is demonstrated for $\delta \geq 10^{-2}$. The growing *relative* deviation at
419 small δ is **not** a failure of the law but the roundoff floor of the **autodiff Hessian**: the angular curvature is
420 $K_{\theta\theta} = \delta/r^*$, reconstructed as a difference of $O(\|K\|)$ terms, so its absolute error is $\sim \epsilon\|K\|$ and its relative
421 error $\sim \epsilon/\delta$. This was confirmed directly: on the analytic Mexican hat a **cancellation-free** Hessian returns
422 relative deviation $1.1\text{--}2.2 \times 10^{-16}$ at every $\delta \in [10^{-8}, 0.3]$, while the autodiff Hessian on the same potential
423 returns $2.28 \times 10^{-8} \rightarrow 4.22 \times 10^{-15}$ across that range, exactly $\propto 1/\delta$. **Do not quote “ 2.2×10^{-16} relative”**
424 **for this experiment.** The same ϵ/δ floor applies to every μ^2 probe in this paper; all other sweeps here use
425 $\delta \geq 10^{-4}$, where the floor is $\sim 10^{-12}$ and the quoted relative agreements are sound.

426 **C.4 Σ is genuinely measured, three ways; it runs, and the angular tilt cannot see it.** $\Sigma_{\text{geom}} = r^*(\delta)$; $\Sigma_{\text{HF}} =$
427 $u \cdot q^*(\delta)$ by Hellmann–Feynman, agreeing to $\max|\Sigma_{\text{HF}} - \Sigma_{\text{geom}}| = 0.0$ exactly; and $\Sigma_{\text{FD}} = -dE_{\text{vac}}/d\delta$
428 by central finite difference at $h=10^{-6}$ — the envelope-theorem definition, structurally independent of the
429 Hessian — agreeing to $\max|\Sigma_{\text{FD}}/\Sigma_{\text{geom}} - 1| = 9.2 \times 10^{-11}$. Under the linear spurion Σ runs by +16.1%
430 to +210.1% across the δ grid (anchored-3000-epoch models +19.8/+17.4/+16.1%; designed-150-epoch
431 models +210.1/+169.3/+35.9/+34.4/+41.1%), while under the angular tilt on the *same* checkpoints it
432 does not move at all ($\leq 2.22 \times 10^{-15}$). With u at channel angles $\{0.0, 0.7, -2.3\}$ on one designed checkpoint
433 at $\delta=10^{-2}$, the μ^2 relative spread is 6.21×10^{-15} , as exact channel invariance requires.

434 **C.5 The leading low-energy constant is saturated by the radial resonance.** With $\mu_{\text{LO}}^2 \equiv \delta \Sigma(0)/F^2(0)$,
435 the theory predicts a *relative* leading-order error $x \equiv \delta/(M_{\text{ch}} \mu_{\text{rad}}^2 f)$. Measured: the ratio is 0.99999607,
436 the mean over 8 checkpoints at $\delta \leq 10^{-5}$, maximum deviation 2.20×10^{-5} ; from the measured μ^2 over
437 $\delta \in [10^{-4}, 10^{-2}]$ it lies in $[0.97917, 0.99992]$; and corrections are first order, $d(\text{ratio})/dx = -1.0504$ at small
438 x , i.e. ratio $= 1 - O(x)$ — exactly what ChPT demands of a *leading* low-energy constant. The CLU therefore
439 realizes ChPT’s **resonance saturation of low-energy constants** exactly rather than phenomenologically, the
440 radial (σ) mode being the saturating resonance. **Breakdown is visible and expected:** at $\delta = 0.3$ the softest-
441 radial seed, $\mu_{\text{rad}}^2 = 0.670$, has $x = 0.68$ — not small — and Σ has run +210%; the expansion is uncontrolled
442 there, so $\delta = 0.3$ **must not be quoted for the next-to-leading-order claim**: cap at $x < 0.25$ or state x
443 explicitly.

444 **C.6 Honest scope.** This appendix is tree-level and classical: no loops, chiral logarithms, running low-energy
445 constants or anomalies; “resonance saturation” is an exact algebraic statement about a two-mode classical
446 potential, not a phenomenological fit. It is probe-only on already-trained checkpoints, spurion off by default so
447 as to be bit-compatible with every result elsewhere, on one architecture family — designed exact $SO(2)$, dim 4,
448 laptop-CPU. The emergent MLP arm is untested and would carry its own self-breaking δ_{eff} , so GMOR there
449 should be tested as $\delta \rightarrow \delta + \delta_{\text{eff}}$: an open falsifiable, not a result.